

tests/dynamic/analyzers/ dns_no_plaintext_53_analyzer.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:**
Active self-validated analyzer (§11.4.107(10)). Signal 5 of 6 for the
dynamic data plane — DNS privacy / no plaintext :53 leak. **Fresh
oracle** — no evidence.sh helper exists for this signal, so the parse
is local (awk).

Companion (§11.4.18) to the in-source documentation block
at the top of the script.

Overview

Signal 5: **no plaintext :53 DNS leak** (§11.4.69
network_connectivity; design §11 © DoH/DoT + leak prevention;
§13). Given a tcpdump capture taken **on the real uplink**, it PASSes
iff **zero** plaintext UDP/TCP :53 DNS packets reach a NON-allowed
resolver. Legitimate DNS under dynamic mode is DoH/DoT
(443/853, encrypted); a plaintext :53 packet on the real uplink to
an external resolver is a DNS leak. An optional allow-list (the in-
tunnel resolver IP[s]) excludes the sanctioned in-namespace
resolver.

Prerequisites

- Source library tests/dynamic/lib/analyzer_common.sh. (This analyzer parses locally with awk; it uses ab_pass_with_evidence from evidence.sh when available, else falls back to ac_pass.)
- awk, grep, POSIX sh.
- A real-uplink tcpdump capture (live P10 input); the self-test feeds bundled fixtures with no network.

Usage examples

```
# Analyze a capture; allow-list (CSV) of sanctioned in-tunnel  
  resolvers optional:  
tests/dynamic/analyzers/dns_no_plaintext_53_analyzer.sh analyze  
  <capture> [allow-csv]
```

```
# Self-validate (golden-good PASS + golden-bad FAIL) – the default action:
```

```
tests/dynamic/analyzers/dns_no_plaintext_53_analyzer.sh --selftest
```

The allow-list may also be supplied via `HELIX_DNS_ALLOW_RESOLVERS` (CSV).

Edge cases

- **Capture file missing** → FAIL.
- **Default allow-list is EMPTY (fail-closed)** — any plaintext :53 is a leak unless an explicit sanctioned resolver is allow-listed (§11.4.6 no-guessing: a leak is never assumed benign). A :53 packet to an in-tunnel resolver that is **not** allow-listed still FAILs.
- **:53 only to an allow-listed in-tunnel resolver** → PASS.
- **DoH-only capture (zero :53 on the uplink)** → PASS.

§11.4.115 RED_MODE polarity

Consuming suites run the §11.4.115 polarity citing this analyzer; RED reproduces a plaintext :53 leak, GREEN guards zero-leak (DoH/DoT or allow-listed resolver only).

Internal behaviour

- `#!/usr/bin/env bash`, POSIX-clean (`sh -n + bash -n`, §11.4.67); the parse is done by `awk`.
- `_dns_leaked_resolvers` walks each capture line, converts `ip.port`: tokens, keeps only `port == 53`, and prints each non-allowed resolver IP (destination for queries, source for responses).
- A non-empty leak list → FAIL with the first offender + count; otherwise `ab_pass_with_evidence` (or `ac_pass` when `evidence.sh` is absent).
- Self-test asserts: DoH-only PASS, allow-listed resolver PASS; plaintext to 1.1.1.1 FAIL, in-tunnel-but-not-allow-listed FAIL; missing-capture negative → FAIL.

Related

- `tests/dynamic/lib/analyzer_common.sh` — sourced base.
- Fixtures: `tests/dynamic/analyzers/fixtures/dns_no_plaintext_53/`.

- Constitution §11.4.69 / §11.4.107 / §11.4.115 / §11.4.6; design §11 ② / §13.

Last verified

2026-07-01 — self-test PASS (DoH-only + allow-listed PASS; plaintext-to-1.1.1.1 + not-allow-listed FAIL); sh -n + bash -n parse-clean. Live real-uplink capture is exercised in **P10**.